

BRD – Water Distribution Inventory & Sales Management System

08/10/2026 · Amritha · Version 4.0 – updated after client feedback on the prototype

1. Purpose

This document sets out the agreed business requirements for a web-based system to manage bottle inventory, production (filling), sales, pricing, quotations, running expenses, billing and payments for the client's water distribution business. Version 4.0 adds the changes the client asked for after reviewing the prototype on 08/10/2026. It is the baseline for the System Architecture & Design document and the Master Task Breakdown.

Item	Detail
Document	Business Requirements Document (BRD)
Version	4.0 (final – updated after client feedback on 08/10/2026)
Previous version	3.1 (final, after prototype review – 04/10/2026)
Prepared by	Amritha
Prepared for	Client – Thuhina Water (water distribution company)
Related documents	Clickable prototype (/prototype); System Architecture & Design (<code>System_Architecture_and_Design.md</code>); Master Task Breakdown (<code>Master_Task_Breakdown.md</code>); Technology Stack & Hosting (<code>TechStack_Thuhina_Water.md</code>)
Status	Final – ready for build

Changes in version 4.0

Requirement tags used in this document: **NEW** added in v4.0 · **UPDATED** changed in v4.0 · **REMOVED** dropped in an earlier version.

#	Change	Sections affected
1	Agreed customer prices. Prices are not the same for every customer: the business agrees a rate with each customer depending on their needs (for example a factory taking 3,500 bottles a month pays less per bottle than one taking 50). The standard price list by bottle type and customer type stays as the default; an agreed price per bottle type can be set for any customer, with a start date, an optional end date and a reason. Bills always keep the price used on the day.	2, 3, 4, 6.1 (FR-05, FR-07), 6.2 (FR-08, FR-11, FR-12), 6.6, 6.9 (FR-53), 6.13 (FR-66), 7 (BR-08, BR-18), 8, 10
2	Customer Active / Inactive status on the customer screen, with the date and a reason; a Deactivate / Reactivate button on every customer in the list.	6.2 (FR-13)
3	New Expenses page: running expenses entered month by month (electricity, water bill, salaries, rent, vehicle fuel, vehicle spare parts and more). Profit report shows net profit after expenses.	4, 5, 6.12 (FR-62, FR-63), 7 (BR-19), 8, 12
4	Send empties: one entry sends every bottle type to the factory at once, each with its own quantity, under one dispatch number, with a printable dispatch note.	6.4 (FR-23), 6.11 (FR-60), 12
5	Customer quotations: price quotations to customers and new organisations (government offices, medium-size factories, hotels, hospitals) – print, email, accepted / not accepted; an accepted quotation registers the customer and sets their agreed prices.	4, 5, 6.13 (FR-64 – FR-66), 7 (BR-20), 8, 12
6	Document numbers added for dispatches, expenses and customer quotations; supplier quotations renamed SQ- .	6.11 (FR-60)

Earlier changes (versions 3.0 and 3.1)

Customer registration records the bottles given that day (first purchase in one step) · first delivery = the very next delivery day · all codes and numbers generated by the system (the paper bill number is kept only for the duplicate check) · more than one filling factory · payment terms and supplier items chosen from fixed lists · purchase orders in fixed steps with only the next step offered · Delivery Planning page with load sheet · Enter Bills lists the customers due automatically · every bill printable, calendar picker on every date · no email server in Phase 1, WhatsApp Option A included · role access and application structure aligned with the prototype. *(Version 3.0 had removed special customer prices; version 4.0 brings them back as agreed customer prices – change 1.)*

All prices in this document are examples, except the Rs. 1,000 deposit for a 20L bottle. Actual prices are set up in the system and can be changed.

2. Business background

The client distributes sealed drinking water in reusable bottles (mainly 20L, also 10L and other sizes) to about 3,000–5,000 registered customers – households, shops, offices, factories and, increasingly, government offices and other organisations. It also sells related products such as water dispensers, stands and pumps.

The business runs on a bottle-exchange model:

- The company buys new empty bottles from suppliers, using quotations and purchase orders. It owns a shared pool of these bottles.
- External filling factories fill and seal the bottles, and charge a fixed price per bottle filled.
- A new customer pays a bottle deposit of Rs. 1,000 for each bottle. This is not refundable. A customer who has an old American Water bottle can hand it in instead of paying the deposit for that bottle.
- After that, on each delivery the customer hands back empty bottles and receives the same number of sealed, filled bottles. They pay only the water price.
- Bottles rotate through the pool, so a customer never gets back the exact bottle they returned.

Water prices. There is a standard price list by bottle type and customer type. Many customers, especially large ones, have an **agreed price** that depends on how much they take and what was negotiated. Example: a factory taking 50 bottles a month pays the standard Rs. 600 per bottle, while a factory taking 3,500 bottles a month has an agreed price of Rs. 475. Larger organisations usually ask for a written **quotation** before they start.

Example: bottle deposit Rs. 1,000 + water Rs. 350 (example price) = Rs. 1,350 for the first bottle, then Rs. 350 per bottle on each exchange. A customer who hands in an old American Water bottle pays only Rs. 350 for that bottle.

The business also pays **running expenses** every month – electricity, water bill, salaries, rent, vehicle fuel and spare parts – which the owner wants to record to see the real profit.

3. Business process UPDATED

Bottle supplier (quotation and PO) → **Empty stock** (company store) → **Filling factories** (fill and seal) → **Filled stock** (ready for delivery) → **Customer** (pays the water price only – their agreed price or the standard price)

Empties are collected on the same visit and go back to empty stock. Damaged or lost bottles are written off from the stage where they are found. The same pool of bottles moves in a loop. Bottles are added by purchases and by old American Water bottles handed in by customers. They leave only when written off.

1. **Buy empties:** The company requests quotations from suppliers (suppliers of the chosen items are suggested) and compares them manually. It creates a purchase order (PO) from the chosen quotation, approves it and sends it. When the goods arrive, they are received against the PO and added to empty stock.
2. **Fill:** Empties of every bottle type are sent to a chosen filling factory in one dispatch, each type with its own quantity. The factory returns them filled and sealed, and charges a fixed price per bottle filled. UPDATED

3. **Quote (larger customers):** A government office, factory or other organisation asks for a price. The company sends a quotation (printed or emailed). When it is accepted, the organisation is registered as a customer and the quoted prices become its agreed prices. **NEW**
4. **Register customer and first purchase:** A new customer is registered with a customer type, area (route), delivery day, cycle (for example every 2 weeks), payment type (Cash, Credit or Monthly bill) and, if agreed, an agreed water price. The filled bottles given on the registration date are recorded in the same step: Rs. 1,000 deposit for each bottle (not refundable) plus the water price; each old American Water bottle handed in waives one deposit and joins the company pool. A bill can be printed for the customer. The first delivery is set to the very next delivery day.
5. **Plan the next day:** In the evening, the manager opens Delivery Planning for tomorrow: customers due by route, bottles needed versus filled bottles ready, and a printable load sheet.
6. **Scheduled exchange:** On the customer's delivery day, staff deliver filled bottles, collect the empties and issue a paper bill. The customer pays the water price only.
7. **Customer not home:** Staff phone the customer. With the customer's agreement, they leave the filled bottles at the door and collect the empties on the next trip. The bill is marked "Left at door" with a note.
8. **Extra orders:** A customer can order more bottles on any day. Any bottles beyond their current holding need a new deposit (or an accepted old bottle).
9. **Enter bills:** Staff hand the paper bills to the Admin. The Admin selects the bill date; the customers due that day are listed automatically, and the Admin types in each paper bill. The system uses each customer's price for that date. Stock, customer bottle balances, money owed and the next delivery date update automatically. Each bill gets a system bill number.
10. **Refill cycle:** Collected empties go back to empty stock and on to the factory. For example, empties collected during the week are filled and redistributed from Sunday.
11. **Damage:** Bottles damaged by the company are disposed of and written off at the company's cost. If a customer damages a bottle, the customer is responsible. To keep getting water for it, they must pay for a new bottle; otherwise their holding is reduced.
12. **Shortages:** If a customer returns fewer empties than they receive, for any other reason, the difference is recorded as bottles owed, with a note. It must be settled later.
13. **Billing and collection:** Cash customers pay on delivery. Credit customers pay later, within their credit limit. Monthly bill customers receive one invoice per month. Payments are recorded against the customer's account.
14. **Price changes:** The standard price list can be changed from a date. A customer's agreed price can be changed, given an end date or removed (back to standard) from a date. Bills already saved keep their price. **NEW**
15. **Customer stops buying:** The customer is made inactive with the date and reason. They drop off the delivery lists; their history, bottles and balance stay on the account and they can be reactivated. **UPDATED**
16. **Running expenses:** Each month the office records the bills paid – electricity, water, salaries, rent, fuel, vehicle parts and others – against the month they are for. **NEW**

4. Scope

Phase 1 is a web application for office use. The admin enters delivery and payment data from paper bills.

In scope (Phase 1)

- Product and bottle-type setup, with a standard price list by bottle size and customer type
- **Agreed water prices for individual customers, with dates and history** **NEW**
- Customer management, including delivery day, cycle, payment type, bottles given at registration and active / inactive status
- **Customer quotations** for customers and new organisations **NEW**
- Purchasing: supplier quotations, purchase orders (POs) and receiving goods against a PO
- Production: one or more filling factories; sending empties of all bottle types in one dispatch and receiving filled bottles, with a fixed charge per bottle **UPDATED**

- Inventory of bottles by status, plus other products
- Bottle deposits, including accepting old American Water bottles in place of the deposit
- Sales: first purchases, scheduled exchanges, extra orders and product sales
- Recording "left at door" deliveries and empties to collect on the next trip
- Tracking the bottle balance each customer holds or owes
- Damage handling: write-off for company damage, and charging the customer for customer damage
- Billing and payments: Cash, Credit and Monthly bill; printable bills, invoices, statements and receipts
- Daily delivery lists and next-day delivery planning with load sheet
- **Running expenses by month, and net profit** NEW
- Importing existing customer data (including agreed prices) and opening stock (one-time, before go-live)
- Reports and dashboard
- User roles: Admin, Accountant, Delivery Staff
- WhatsApp Business messages – essential messages (Option A, see FR-61)

Out of scope (Phase 1)

- Mobile app (planned for a later phase)
- Route, vehicle and driver management (customer area is used as the route)
- Online ordering or a customer portal
- Payment gateway integration
- Management of the filling factories' own operations
- Automatic price bands by monthly volume (agreed prices are set by the Admin – see section 10)
- Sending email from the server – quotations are emailed from the user's own email program; see section 11
- Full accounting (general ledger, VAT returns, payroll) – only running expenses are recorded

5. Users and roles UPDATED

Three roles use the system through a web browser. Each role sees only the menus and functions listed for it. There is no public signup – users are created by the Admin.

Role	Main responsibilities	Access
Admin	Sets up products, standard prices, customers and their agreed prices, suppliers and factories. Sends customer quotations. Records supplier quotations, creates and approves purchase orders, and receives goods. Enters factory dispatches, deliveries and cash payments from paper bills. Plans next-day deliveries. Records damage, write-offs and expenses, and manages users.	Full access to all menus
Accountant	Manages Credit and Monthly bill customers, invoices and payments. Records running expenses. Reviews outstanding balances, supplier and factory amounts owed, costs and profit.	Dashboard; Billing & Payments (full); Expenses (full) ; Reports – Sales, Purchasing, Outstanding & aging, Profit, Expenses ; view only: Customers (incl. prices), Suppliers & Factories, Purchasing, Filling Factory, Customer Quotations
Delivery Staff	Views and prints the daily delivery list, including customer bottle balances and empties to collect.	Daily Delivery List (view and print)

Only the **Admin** can set or change a customer's agreed price.

6. Functional requirements

Requirements are grouped by module. Each has an ID so it can be traced through design and testing. IDs are never reused or renumbered.

6.1 Products and pricing

FR-01 Maintain bottle types (for example 20L and 10L), each with a name, size in litres and active/inactive status. The bottle-type code is generated from the size (for example B20).

FR-02 Maintain other products (dispensers, stands, pumps, etc.) with a selling price, cost price and stock quantity. The product code is generated by the system.

FR-03 Maintain customer types (for example Household, Shop, Office, Factory).

FR-04 Set the **standard** water price by bottle type and customer type. Set the bottle deposit by bottle type (currently Rs. 1,000 for a new 20L bottle).

FR-05 **UPDATED** Set an **agreed water price** for an individual customer and bottle type, which overrides the standard price for that customer only:

- agreed price per bottle, **from** date (can be in the future – scheduled), optional **until** date (for example the end of a tender or trial), and a **reason** (for example "Volume price – about 3,500 bottles a month");
- set at registration, from an accepted quotation (FR-66), from the customer's details panel, or by import (FR-53);
- changing the price closes the previous agreement the day before; "back to standard" ends it from a date;
- shown with the standard price, the difference, the bottles a month at the customer's usual quantity and the filling charge as a guide;
- full price history per customer, and a list of all customers on agreed prices.

FR-06 Maintain a list of accepted old-bottle brands that can be handed in instead of paying the deposit, and the bottle size each is accepted for. The list starts with American Water (20L), and more can be added later.

FR-07 **UPDATED** Keep a price history with effective dates for standard prices, deposits and agreed customer prices. A price change can be scheduled for a future date. Every bill stores the unit price used, so past bills keep the price that applied at the time.

6.2 Customer management

FR-08 Register customers with name, address, phone, customer type, area (route) and registration date. The customer code is generated by the system. On registration, record any agreed water price, the filled bottles given that day, any old American Water bottles handed in and the cash paid; this creates the first-purchase bill (FR-32) at the customer's price.

FR-09 Set each customer's delivery day, cycle (every 1, 2, 3 or 4 weeks) and usual quantity per bottle type. The first delivery date is filled in automatically as the very next delivery day after the registration date and can be changed with the calendar.

FR-10 Set each customer's payment type: Cash, Credit or Monthly bill. Credit and Monthly bill customers get a credit limit and payment terms chosen from a fixed list (7, 14, 30, 45 or 60 days).

FR-11 **UPDATED** Show each customer's bottles held (by bottle type), empties waiting to be collected, bottles owed, money outstanding, credit limit, delivery details, **water price (agreed or standard, with the standard shown)** and full history (bills, payments, invoices and **price history**).

FR-12 **UPDATED** Search and filter customers by name, phone, code, area, delivery day, type, payment type, status and **price (standard / agreed)**.

FR-13 **UPDATED** Make a customer **Inactive** without deleting their history, from the customer's edit screen (Status: Active / Inactive) or with the Deactivate button on the customer list:

- "Inactive from" date (not in the future) and a reason from a list (Stopped buying, Moved away, Business closed, Went to another supplier, Not paying, Other – Other needs details);
- a warning if the customer still holds bottles, has empties to collect, owes bottles or owes money – these stay on the account;
- inactive customers are not listed for delivery and cannot get new bills; payments can still be received;
- a customer can be **reactivated** at any time; the next delivery date moves to the next delivery day from today;

- the list shows the status with "since dd/mm/yyyy" and the counts of active and inactive customers.

6.3 Purchasing (quotations and purchase orders)

FR-14 Maintain suppliers of empty bottles and resale products. The items each supplier supplies are chosen from the system's item list (bottle types and products). Payment terms are chosen from a fixed list (Cash on delivery, 7, 14, 30, 45 or 60 days). The supplier code is generated by the system.

FR-15 Create a quotation request listing the items and quantities needed, and the suppliers asked to quote. Suppliers who supply the chosen items are ticked automatically. The request number is generated by the system.

FR-16 Record each supplier's quotation: unit prices, delivery time, validity date, the supplier's own reference and notes. Allow the scanned quotation to be attached.

FR-17 Show all quotations for the same request side by side, with the lowest prices highlighted for information. The owner chooses the quotation manually and adds a reason. The system does not choose automatically. Expired quotations cannot be chosen.

FR-18 Create a purchase order (PO) from the chosen quotation, copying its items and prices. Allow edits while the PO is a draft. Each PO gets a unique number and can be printed or saved as PDF.

FR-19 Track PO status in fixed steps: Draft → Approved → Sent → Partly received / Received, or Cancelled (from Draft or Approved, with a reason). The screen shows the current step and offers only the next step's action.

FR-20 Receive goods against a sent PO, recording the quantities received and any damaged on arrival. Good bottles are added to empty stock and products to product stock. Damaged items are not added to stock and are not paid for.

FR-21 Track amounts owed to suppliers (value of good items received) and payments made (cash, bank transfer or cheque).

6.4 Production (filling factories)

FR-22 Maintain one or more filling factories, each with contact details, payment terms (fixed list), active/inactive status and a fixed filling charge per bottle, set by bottle type. The factory code is generated by the system.

FR-23 **UPDATED** Record a dispatch to a chosen factory in **one entry for all bottle types**: date, vehicle, note, and the number of empty bottles sent for **each** bottle type (default = all empties in store; 0 = not sent). All lines are checked together (enough empty stock, the factory has a charge for each type) and saved under one **dispatch number**; one batch is created per bottle type and the charge in force at dispatch is kept with each batch. A **dispatch note** listing every bottle type can be printed.

FR-24 Record the return from the factory against each batch: date, quantity of filled bottles received and quantity rejected or damaged (with a reason). Rejected bottles are written off as company damage. A batch can be returned in more than one part.

FR-25 Show bottles currently at each factory, and any difference between bottles sent and received for each batch.

FR-26 Calculate the filling cost for each batch (bottles filled × charge per bottle) and track amounts owed to, and payments made to, each factory by month.

6.5 Inventory

FR-27 Track every bottle type across five statuses: Empty in store, At factory, Filled in store, With customers and Written off.

FR-28 Update stock automatically from goods receipts, factory dispatches and returns, deliveries, collections, old bottles handed in and write-offs. Keep a movement history.

FR-29 Record each damaged bottle with its responsibility:

- Company (empty in store, filled in store, at the factory or during delivery): the bottle is disposed of and written off as a company cost.

- Customer: the customer is responsible and is charged for a new bottle (FR-37).

FR-30 Record lost bottles and stock adjustments after a physical count, with a reason and the user who made them.

FR-31 Alert when filled stock falls below a set minimum level. The minimum level is set per bottle type.

6.6 Sales and deliveries

All water prices in this section are the **customer's price on the bill date**: the agreed price in force, otherwise the standard price for the customer's type (FR-04, FR-05).

FR-32 Record a first purchase: bottle deposit + water price for each bottle. Add the bottles to the customer's holding. This is normally done during registration (FR-08) and can also be done later for a registered customer with no bottles.

FR-33 In a first purchase or extra order, record any old American Water bottles handed in. Each one waives the deposit for one bottle and is added to empty stock.

FR-34 **UPDATED** Record scheduled exchanges from paper bills. When the Admin selects the bill date, the customers due that day are listed automatically with their usual quantities. For each bill record: paper bill number, filled bottles given, empties collected, cash paid and a note. The amount uses the customer's price, and an agreed price is marked on the row. The system gives each bill its own bill number.

FR-35 Record a "Left at door" delivery when the customer is not home: confirm that the customer agreed by phone, add a note and record the filled bottles left. The empties are recorded as to collect on the next trip. This is not counted as a shortage.

FR-36 When the empties are collected on the next trip, clear them from the customer's "to collect" balance – either automatically when extra empties are entered on the next bill, or with a separate "Collect empties" entry.

FR-37 Record a bottle damaged by the customer. The damaged bottle is written off. Either the customer pays the price of a new bottle (the deposit) and receives a filled replacement (water charged), keeping the same holding, or the bottle is not replaced and the customer's holding is reduced.

FR-38 For any other case where a customer returns fewer empties than they receive, record the difference as bottles owed, with a note. It is settled later by returning bottles or paying the deposit (the customer then keeps the bottle).

FR-39 Record extra orders on any day, in any quantity. Bottles beyond the customer's current holding need a deposit (or an accepted old bottle).

FR-40 Record sales of other products (dispensers, stands, pumps) to registered customers or walk-in customers (walk-in sales are paid in full).

FR-41 Search for a bill by its system bill number or paper bill number, to prevent duplicate entry. A saved bill can be viewed, printed or reversed (BR-14).

6.7 Delivery schedule

FR-42 Produce a daily delivery list of customers due that day, based on delivery day and cycle, including customers whose last delivery was missed. Inactive customers are not listed.

FR-43 For each customer on the list, show address, phone, payment type, bottles held, usual quantity, empties left from the last trip, bottles owed and money outstanding.

FR-44 Print the delivery list.

FR-45 Show the filled bottles needed for upcoming days, to help plan factory batches.

FR-46 After a delivery is entered, move the customer's next delivery date forward by their cycle.

FR-58 Delivery Planning (Admin): choose any date from the calendar (default tomorrow; quick buttons Today, Tomorrow, Next week) and an area (route). Show:

- customers due on that date, or all customers, grouped by area (route) with bottles to load per route;

- for each customer: last delivery made (date and bottles), quantity for the selected date, next delivery and the delivery after that, empties to collect, bottles owed, balance and status (due, entered, missed, first delivery, over credit limit);
- bottles needed per bottle type versus filled bottles ready (allowing for bottles still needed before that date), with a clear "Ready" or "Short by N" and the number at the factories;
- a printable load sheet: totals per route and the customer list for the drivers.

6.8 Billing and payments

The system supports three payment types:

Payment type	How it works
Cash	The customer pays on delivery. The Admin records the payment from the paper bill. Any amount not paid stays on the customer's balance.
Credit	Each sale is added to the customer's balance, within their credit limit. The customer pays later.
Monthly bill	Deliveries are added up and invoiced at the end of each month.

FR-47 Record cash payments collected on delivery.

FR-48 For Credit customers, add each sale to the account balance and check it against the credit limit. Show balances against limits and flag customers over their limit.

FR-49 Generate monthly invoices for Monthly bill customers, listing all deliveries in the period (with the unit price of each), the previous balance, payments received and the total due.

FR-50 Record payments received against customer balances, including part-payments, and how the money was received (cash, bank transfer or cheque, with the reference). A receipt can be printed.

FR-51 Show an aging report of outstanding balances (0–30, 31–60, 61–90 and 90+ days). Payments settle the oldest amounts first.

FR-52 Print or export customer statements, invoices and receipts as PDF.

FR-59 Print any bill (first purchase, exchange, extra order, product sale, customer-damaged bottle), including the first bill given at registration.

6.9 Data migration

FR-53 **UPDATED** Import existing customers from Excel/CSV, including payment type, bottles held, empties to collect, bottles owed, money outstanding and, optionally, an agreed price per bottle type (blank = standard price).

FR-54 Import opening stock by bottle type and status.

FR-55 Validate imported data and show a report of rejected rows, with reasons, before confirming the import.

The data migration screen is hidden in the prototype for now. It remains in scope and is enabled for the one-time import before go-live.

6.10 Users and security

FR-56 Log in with a username and password. Access is limited by role. The Admin creates users, assigns roles, deactivates users and resets passwords (temporary password, changed at first login).

FR-57 Keep an audit log of who created, changed, approved, reversed, deleted or imported records, and when – including every price change. Audit entries cannot be edited or deleted.

6.11 System-wide

FR-60 **UPDATED** All codes and sequential numbers are generated by the system and cannot be typed or changed by users.

Record	Example format	Record	Example format
Customer	C0001	Purchase order	PO-2026-0001
Product	P01	Goods receipt	GRN-0001
Bottle type	B20 (from size)	Factory dispatch NEW	FD-0001
Supplier	S01	Factory batch	FB-0001
Filling factory	F01	Bill	B000001
Quotation request	QR-2026-001	Monthly invoice	INV-2609-0001
Supplier quotation UPDATED	SQ-0001	Payment receipt	RC-00001
Customer quotation NEW	QT-2026-001	Expense NEW	EX-0001

FR-61 WhatsApp Business messages (Phase 1 – Option A, essential messages). Send messages through the WhatsApp Business Cloud API to customers who have agreed to receive them: "Left at door" notice (when the bill is saved), payment reminder (Credit and Monthly bill customers with an overdue balance) and monthly invoice (PDF). Each message type can be switched on or off in settings, and every message sent is logged. Thuhina Water owns the WhatsApp Business account and pays Meta per message. Delivery reminders before every delivery (Option B) are not in Phase 1 and can be added later.

6.12 Running expenses **NEW**

FR-62 Record running expenses: date paid, the **month the expense is for** (for example September electricity paid on 2 October counts for September), expense type, description, paid to, bill or account number, amount, and how it was paid (cash, bank transfer or cheque with the reference). Expenses can be edited; deleting needs a reason and is kept in the audit log.

FR-63 Maintain expense types (starting with Electricity, Water bill, Salaries, Rent, Vehicle fuel, Vehicle spare parts & repairs, Telephone & internet, Stationery & printing, Other; more can be added and unused ones deactivated). Show expenses month by month: total for the month, the previous month and the difference, totals by type, a month-by-month table by type, a printable month statement and Excel export.

6.13 Customer quotations **NEW**

FR-64 Create a quotation to an **existing customer** or to a **new organisation** (name, contact person, designation, address, area, phone, email). Choose the customer type used for prices. Record the date, valid-until date (default 30 days), subject (for example a tender reference), delivery arrangement, payment terms (fixed list) and terms and conditions (default text, editable). Lines can be water per bottle type, bottle deposit per bottle type, products or free text, each with description, quantity, unit and unit price. Unit prices start at the customer's current price (agreed or standard) or, for a new organisation, the standard price for the chosen type; they can be changed, and the current price is shown beside a changed price.

FR-65 Track the quotation in fixed steps: **Draft** → **Sent** (by email, by hand, post or WhatsApp, with the date) → **Accepted** or **Not accepted** (with a reason). A sent quotation past its valid-until date shows **Expired** and can still be answered. Only a draft can be edited; "Make a copy" starts a new draft. Every quotation can be printed or saved as PDF (company header, customer, subject, lines, total, delivery, terms, signatures) and emailed (the email program opens with the message ready; the PDF is attached by the user).

FR-66 When a quotation is **accepted**: a new organisation can be registered as a customer in one step, with its details and the quoted water prices filled in; for an existing customer, "Apply prices" makes the quoted water prices the customer's agreed prices from a date (FR-05). The quotation keeps the link to the customer and to the prices it set.

7. Business rules

ID	Rule
BR-01	A customer pays a Rs. 1,000 deposit for each new bottle before receiving water in it. The deposit is not refundable.
BR-02	An old American Water bottle handed in replaces the deposit for one bottle. The bottle joins the company pool and is filled like any other.
BR-03	On an exchange, the customer pays only the water price for each filled bottle.
BR-04	A "Left at door" delivery needs the customer's agreement by phone and a note. Its empties are collected on the next trip and are not treated as a shortage.
BR-05	Bottles damaged by the company are disposed of and written off at the company's cost.
BR-06	Bottles damaged by the customer are the customer's responsibility. The customer must pay for a new bottle to keep getting water for it; otherwise their holding is reduced.
BR-07	Any other shortage (filled given minus empties returned) is recorded as bottles owed, and stays on the account until settled.
BR-08 UPDATED	The water price on a bill is the customer's agreed price for that bottle type in force on the bill date; if there is none, the standard price for the bottle type and customer type.
BR-09	Every customer has one payment type: Cash, Credit or Monthly bill. A Credit or Monthly bill customer cannot exceed their credit limit without Admin approval, which is recorded in the audit log.
BR-10	A purchase order is created only from a quotation chosen by the owner. It must be approved and then sent before goods are received against it.
BR-11	Each filling factory is paid a fixed price per bottle filled (its own rate per bottle type). Rejected bottles are not charged.
BR-12	Stock cannot go below zero. The system blocks the entry and asks for a stock adjustment.
BR-13	Every bill gets a unique system bill number. A paper bill number can be entered only once.
BR-14	Saved transactions are not deleted. Corrections are made by reversal entries, recorded in the audit log. After a bill is reversed, its paper bill number can be entered again correctly.
BR-15	Bottles rotate through the pool. The system tracks quantities by bottle type, not individual bottles.
BR-16	A new customer's first delivery is the very next delivery day after the registration date. After that, deliveries follow the customer's cycle.
BR-17	Codes and document numbers are generated by the system only (FR-60).
BR-18 NEW	Only the Admin sets or changes an agreed price. Every agreed price has a start date and a reason; it may have an end date. Only one agreed price per customer and bottle type is in force on any date. A change never alters bills already saved – each bill keeps the unit price it was saved with. The bottle deposit is the same for every customer.
BR-19 NEW	An expense counts in the month it is for, not the month it was paid. A cheque or bank transfer needs its reference. Deleting an expense needs a reason.
BR-20 NEW	A customer quotation can be changed only while it is a draft. Only the next step is offered (Draft → Sent → Accepted / Not accepted). An accepted quotation changes a customer's prices only when it is registered or "Apply prices" is confirmed.

8. Reports and dashboard UPDATED

All reports filter by date range and can be exported to PDF and Excel.

Report	What it shows	Main users
Dashboard	Today's sales, collections, filled stock (with low-stock alert), bottles at the factories, deliveries due vs entered, empties to collect, bottles owed, outstanding total, items needing attention (incl. quotations not sent, quotations about to expire or accepted but not yet a customer / prices not applied, agreed prices ending within 30 days, last month with no expenses)	Admin, Accountant
Delivery planning	Customers due on a chosen date by route, bottles needed vs ready, load sheet (FR-58)	Admin
Sales	Sales by day/month, customer, customer type, bottle type and sale type (first purchase, exchange, extra order, product); customers on agreed prices: water billed vs the standard price, and the total difference	Admin, Accountant
Purchasing	Quotations by request and supplier, purchase orders and their status, goods received vs ordered	Admin, Accountant
Production	Bottles sent to and received from each factory, rejects, filling cost per batch	Admin
Stock	Stock by bottle type and status at a date, plus product stock	Admin
Bottle movement	Bottles in circulation, with customers, empties to collect, bottles owed, old American Water bottles taken in	Admin
Damage and write-off	Damaged and lost bottles by responsibility (company or customer), location and reason	Admin
Outstanding and aging	Money owed by each customer, by payment type and age	Admin, Accountant
Profit	Sales revenue minus bottle, product and filling costs, by period and bottle type = gross profit; minus running expenses by type = net profit (expenses counted for whole months)	Admin, Accountant
Expenses NEW	Expenses by type and month, all entries, month totals chart	Admin, Accountant
Delivery	Deliveries due vs entered, and "Left at door" deliveries, by day	Admin

9. Non-functional requirements

ID	Area	Requirement
NFR-01	Platform	Web application that works in current Chrome, Edge and Firefox on desktop and laptop
NFR-02	Capacity	Supports at least 10,000 customers and several hundred bill entries per day without slowing down
NFR-03	Speed	Common screens and searches load within 3 seconds
NFR-04	Data entry	Fast keyboard-friendly entry of paper bills: customers due on the bill date are listed automatically, Enter moves to the next field, one key saves all bills
NFR-05	Security	Role-based access, encrypted (hashed) passwords, HTTPS
NFR-06	Backup	Automatic daily backup with a copy kept off the server, and a restore test every month
NFR-07	Currency, dates	Sri Lankan Rupees (LKR); dates shown as DD/MM/YYYY and selected with a calendar picker (typing also allowed)
NFR-08	Future-ready	Design allows a mobile app for delivery staff to be added later

ID	Area	Requirement
NFR-09	Usability	Simple screens: one page per main activity; add, edit and view in pop-ups or a side panel on the same page; only the next valid action is offered

10. Assumptions and open items

Assumptions

- Delivery staff issue paper bills. The admin enters them into the system afterwards.
- The company may use more than one filling factory.
- Bottles are tracked by quantity and type, not by individual bottle serial numbers.
- The customer's area is used as the delivery route; route, vehicle and driver management is not needed in Phase 1.
- The client provides existing customer data (including any agreed prices) and stock data in Excel for import.
- Agreed prices are decided by the owner case by case; the system records them and shows the standard price, the difference and the filling cost as a guide. It does not calculate prices from volume automatically.
- Hosting, backup and running costs are described in the Technology Stack & Hosting document.

Resolved

- ☒ Customer-specific prices – **needed** (v4.0): agreed price per customer and bottle type, dated, set by the Admin (replaces the v3.0 decision "fixed prices only")
- ☒ Accepted quotation at a lower price – becomes the customer's agreed price (v4.0)
- ☒ Number of filling factories – more than one allowed
- ☒ Customer types – Household, Shop, Office, Factory (more can be added)
- ☒ Payment terms – fixed list (Cash on delivery, 7, 14, 30, 45, 60 days)
- ☒ Areas / zones – customer area is used as the route
- ☒ Email – no email server in Phase 1; quotations emailed from the user's email program
- ☒ Paper bill number – kept on daily bill entry for the duplicate check (BR-13)
- ☒ WhatsApp – included in Phase 1 with essential messages only (Option A)
- ☒ Running expenses – recorded month by month; net profit in the Profit report (v4.0)

Open items for the client to confirm

- ☐ Actual standard water prices for each bottle type and customer type (10L example prices: Household 200, Shop 190, Office 185, Factory 175)
- ☐ List of customers with agreed prices today, with their prices (for the import)
- ☐ Should very large customers ever get a different bottle deposit? (v4.0 keeps one deposit for everyone – BR-18)
- ☐ Would automatic price bands by monthly volume be useful later (for example Factory: up to 500 bottles Rs. 600, 500–2,000 Rs. 525, over 2,000 Rs. 475)? Not in Phase 1.
- ☐ Deposit amount for bottle sizes other than 20L (10L example: Rs. 600)
- ☐ Filling charge per bottle for each factory and bottle type
- ☐ Full list of bottle types, other products, areas and expense types
- ☐ Credit limits for Credit and Monthly bill customers
- ☐ Does any tax (e.g. VAT) apply to invoices or quotations?
- ☐ Bill, invoice, receipt, PO, quotation, dispatch note and load-sheet layout, including logo and company details
- ☐ WhatsApp Business number and Meta account details; confirm Meta's current message rate for Sri Lanka
- ☐ Number of users for each role

11. Communication

Channel	Decision
Email	No email server in Phase 1. Users are created and their passwords reset by the Admin; bills, invoices, statements and quotations are printed or saved as PDF. A quotation's Email button opens the user's own email program with the message ready, and the user attaches the PDF. If the client later wants the system to send emails itself, the client's own email account (SMTP) can be connected – no email server needed.
WhatsApp Business	Included in Phase 1 – Option A, essential messages only ("Left at door", payment reminders, monthly invoices) (FR-61). No monthly subscription – Meta charges per message sent (about 1,000–1,500 messages a month, est. US\$5–8). Delivery reminders (Option B) can be added later. Details are in the Technology Stack & Hosting document.

12. Application structure (from the prototype) UPDATED

Menu	Page	What it covers	Requirements
Dashboard	Dashboard	Today at a glance, low-stock alert, items needing attention	Sec 8, FR-31
Master Data	Bottles & Products	Bottle types, other products, standard prices & deposits, list of customer agreed prices , accepted old bottles, price history	FR-01–07
	Customers	Customer list (status and price filters), register/edit (with agreed price and bottles given at registration, Status), customer details panel (balances, history, Prices tab and popup), customer types	FR-03, 05, 08–13, 32–33
	Suppliers & Factories	Suppliers (items supplied, terms) and filling factories (charges, terms)	FR-14, FR-22
Purchasing	Purchasing	Quotation requests and comparison, purchase orders (steps, print, receive goods), supplier payments	FR-15–21
Production	Filling Factory	Send empties (all bottle types in one dispatch, dispatch note) , receive filled bottles, batches, factory payments	FR-23–26
Inventory	Stock	Stock by status, minimum levels, damage, adjustments, movements	FR-27–31
Sales & Deliveries	Enter Bills	Daily paper-bill entry grid (customers listed by date, customer's price)	FR-34–36, 38, 41, 46–48
	Sales & Bills	All bills (search, print, reverse); first purchase, extra order, product sale, collect empties, customer-damaged bottle, settle owed	FR-32–33, 36–41, 59
	Customer Quotations NEW	Quotations to customers and new organisations; print, email, accepted / not accepted; register customer and apply prices	FR-64–66
Delivery Schedule	Delivery Planning	Next-day planning by route, bottles ready check, load sheet (Admin)	FR-58
	Daily Delivery List	Daily list and print, upcoming bottle requirement	FR-42–46
Billing & Payments	Billing & Payments	Customer accounts and aging, receive payment, statements, monthly invoices	FR-47–52
Expenses NEW	Expenses	Running expenses by month, expense types, month-by-month summary, print, Excel	FR-62–63
Reports	Reports	Ten reports with date range, PDF and Excel export	Sec 8
Administration	Administration	Users, roles & access, audit log	FR-56–57
Data Migration	Data Migration	Customer (incl. agreed prices) and opening-stock import (hidden until go-live)	FR-53–55

13. Sign-off

Name	Role	Signature	Date
	Client representative		
Amritha	Developer / Business analyst		